

Igor Janotti

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SUMMARY

Software Engineer with experience in systems-level development, distributed systems, and networked services. Strong foundation in data structures, operating systems, and concurrent programming. Experienced building and debugging event-driven and device-oriented systems. Interested in low-level infrastructure, energy systems, and high-performance distributed services.

TECHNICAL SKILLS

Languages: C/C++, Python, C#, Java

Systems: Distributed Systems, Concurrent Programming, Operating Systems Concepts, Event-Driven Architecture

Networking: TCP/IP, Networked Services, Device Communication Protocols (BACnet)

Cloud & Infrastructure: AWS (Lambda, DynamoDB, S3, SNS), Docker

Databases: MySQL, MongoDB

Tools: Git, Linux, GDB (basic), Node.js

Core: Data Structures, Algorithms, Debugging, Performance Optimization

EXPERIENCE

Pacific Northwest National Laboratory

Jun 2025 – May 2026

Software Engineer Intern

Bellevue, WA

- Contributed to a distributed agent-based system for managing real-world energy and building infrastructure
- Developed BACnet device discovery and scanning system to improve visibility of networked building devices
- Implemented networked services enabling communication across distributed system components
- Built event-driven backend features focused on reliability, scalability, and fault tolerance
- Debugged and resolved system-level issues across services, improving runtime stability and performance
- Worked in Linux environments for deployment, testing, and system diagnostics

Bellevue College

Feb 2024 – Mar 2024

Teaching Assistant – Programming Languages (C++)

Bellevue, WA

- Assisted students with debugging C++ programs, focusing on memory management and execution flow
- Reinforced systems concepts including compilation, linking, and runtime behavior

PROJECTS

VOLTTRON AI: Conversational Smart-Grid Control Interface | *Python, Pydantic AI, LLMs*

- Built a model-agnostic conversational interface for VOLTTRON enabling natural language control of smart-grid systems
- Designed AI tool-calling system using Pydantic AI to map user requests to platform operations (agents, drivers, configuration)
- Optimized prompts and tool schemas to support both large cloud models and lightweight local models
- Implemented lazy loading, singleton environment management, and cached path resolution to improve reliability and performance
- Developed modular tool system supporting platform control, configuration management, error analysis, and streaming responses

Cloud-Based License Plate Processing System | *C#, AWS*

- Designed distributed event-driven pipeline using AWS services for real-time data processing
- Built asynchronous workflows for ingestion, processing, and storage across multiple services

AI-Powered CLI Tooling | *Python*

- Developed structured command execution system with validation and deterministic behavior
- Focused on safe execution and predictable system interaction

FollowBot Companion Robot | *C++, Python*

- Implemented real-time navigation and control logic with sensor-driven feedback loops
- Built communication between hardware inputs and system state for responsive behavior

EDUCATION

Bellevue College

Bellevue, WA

Bachelor of Science in Computer Science

Expected Spring 2025

- GPA: 3.54